Comprehensive Networking Notes

1. LAN (Local Area Network)

A **Local Area Network (LAN)** is a network that connects computers and devices in a limited geographic area, such as an office building, campus, or home, typically covering distances up to a few kilometers.

Key Characteristics

- Covers small geographic areas (single building or campus)
- Speed ranges from 10 Mbps to 100 Mbps (modern LANs achieve higher speeds)
- Used to share resources like printers, storage, and internet access
- Low latency and high reliability
- Common topologies: Bus, Ring, and Star

Components of a LAN

- Ethernet cables
- Network switches
- Routers
- Firewalls
- Wireless Access Points (WAPs)
- Desktop and laptop computers
- Printers and IoT devices

Real-Life Example

Your home Wi-Fi network is a LAN. All devices connected to it (computer, phone, smart TV) form a local network and can communicate with each other without going through the internet.

2. IP Address (Internet Protocol Address)

An **IP address** is a unique numerical identifier assigned to every device connected to a network or the internet, enabling communication between devices.

IP Address Format

- Format: Four numbers separated by periods (e.g., 192.158.1.38)
- Each number ranges from 0 to 255
- Total address range: 0.0.0.0 to 255.255.255.255

IP Address Versions

IPv4

- Uses 32-bit addresses
- Supports approximately 4.3 billion unique addresses
- Most commonly used version
- Example: 192.168.1.1

IPv6

- Uses 128-bit addresses
- Supports exponentially more addresses (future-proofing)
- Designed to replace IPv4 as devices multiply
- Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

IP Address Types

Public IP Addresses

- Assigned by Internet Service Providers (ISPs)
- Globally unique and routable on the internet
- Used to identify devices on the public internet

Private IP Addresses

- Used within private networks (LANs)
- Not routable on the public internet
- Common ranges:
    - 10.0.0.0 to 10.255.255.255
    - 172.16.0.0 to 172.31.255.255
    - 192.168.0.0 to 192.168.255.255

Assignment

IP addresses are allocated by the **Internet Assigned Numbers Authority (IANA)**, ensuring no duplicates exist on the internet.

3. Network Switch

A **network switch** is a hardware device that connects multiple devices within a LAN and forwards data packets between them based on MAC addresses.

Key Functions

- Operates at Layer 2 (Data Link Layer) of the OSI Model

- Filters and forwards packets based on MAC addresses
- Performs error checking before forwarding data
- Supports unicast, multicast, and broadcast transmissions
- Uses packet-switching techniques

How a Switch Works

1. Data arrives at a port
2. The destination MAC address is examined
3. Error checking is performed
4. Packet is forwarded only to the device with the matching MAC address
5. Multiple simultaneous connections are supported

Features

- Full-duplex communication mode
- Multiple ports (typically 8, 24, or 48)
- Allocates bandwidth per LAN segment
- Faster than hubs and more efficient than repeaters
- No need for broadcasts to all ports

Benefits

- Efficient bandwidth utilization
- Support for wired connections to printers and IoT devices
- Enables network traffic management
- Better security through segmentation

4. Router

A **router** is a networking device that forwards data packets between different networks, connecting multiple LANs or connecting a LAN to the internet (WAN).

Key Functions

Forwarding

- Receives packets and examines headers
- Forwards them to the correct output port

Routing

- Determines the optimal path for packets

- Uses routing tables and algorithms
- Consults network path information

Network Address Translation (NAT)

- Translates private IPs to public IP
- Enables internet access for internal devices

Security

- Supports firewalls
- Implements security measures

How Routers Work

1. Router receives a packet
2. Reads the destination IP address
3. Consults the routing table
4. Determines the best path to destination
5. Forwards packet to the next router or destination

Routing Types

Static Routing

- Manually configured
- Suitable for small or stable networks
- Routes don't change automatically

Dynamic Routing

- Automatically updated based on network activity
- Ideal for large or changing networks
- Adapts to network changes

5. Gateway

A **network gateway** is a connection point between different networks that enables them to exchange information and acts as a mediator between dissimilar networks.

Key Characteristics

- Functions as an entry or exit point for data
- Translates network protocols
- Can be unidirectional or bidirectional

- Operates at various OSI layers

6. Subnet and Subnetting

A **subnet** (subnetwork) is a logical division of an IP network into smaller networks, and **subnetting** is the process of dividing a large network into multiple subnets.

Subnet Mask

What is it?
A 32-bit number that divides an IP address into:

- Network portion (represented by 1s)
- Host portion (represented by 0s)

Example

- IP: 192.168.1.0
- Subnet Mask: 255.255.255.0
- Network portion: 192.168.1
- Host portion: 0 (allows for 256 addresses, but 254 usable)

Why Use Subnetting?

Problems Solved

- Divides large networks into manageable sub-networks
- Improves network security
- Enhances network efficiency
- Reduces congestion
- Better resource allocation

Example Scenario
A company with 16 million hosts in a Class A network:

- **Without subnetting:** All 16 million hosts in one network → Difficult management
- **With subnetting:** Divide into 4 departmental subnets → Easy management and security

Subnetting Process

1. Identify the number of subnets needed
2. Determine bits to borrow from the host portion
3. Calculate new subnet mask

4. Identify network and broadcast addresses for each subnet
5. Allocate IP addresses to devices

Example: Class C Network Divided into 4 Subnets

- Original: 192.168.1.0/24
- Borrow 2 bits from host portion
- Create 4 subnets:
    - 192.168.1.0/26
    - 192.168.1.64/26
    - 192.168.1.128/26
    - 192.168.1.192/26

Benefits of Subnetting

- Improved security through network segmentation
- Better performance and reduced congestion
- Efficient use of IP addresses
- Easier network management
- Reduced broadcast traffic

7. CIDR (Classless Inter-Domain Routing)

CIDR is a method of allocating IP addresses and routing that allows more efficient use of IP addresses than traditional class-based addressing.

CIDR Notation

Format
a.b.c.d/n

Where:

- a.b.c.d = IP address
- /n = number of bits in the network prefix
- First n bits represent the network
- Remaining bits represent the host ID

Examples

- 192.168.1.0/24 → First 24 bits are network, remaining 8 bits are host
- 10.0.0.0/16 → First 16 bits are network, remaining 16 bits are host

- 172.16.0.0/12 → First 12 bits are network, remaining 20 bits are host

Why CIDR?

Problem: Classful Addressing Wastage

Class	Addresses	Hosts	Problem
A	2^24	2^24 - 2	Too large for small organizations
B	2^16	2^16 - 2	Wastes 49,150 hosts if only 214 needed
C	2^8	2^8 - 2	Too small for large organizations

Solution: CIDR Benefits

- Flexible block allocation
- Matches exact network requirements
- Reduces IP address wastage
- Allows fine-grained control of subnet sizes
- Slows IPv4 address exhaustion

Variable-Length Subnet Masking (VLSM)

- Network prefixes can have variable length
- Efficient allocation of address space
- Enables aggregation of multiple routes into supernets

How to Calculate Usable Addresses in CIDR

Formula: $2^{(32-n)} - 2$

Where:

- n = number of network bits
- Subtract 2 for network address and broadcast address

Example: 192.168.1.0/24

- Host bits: 32 - 24 = 8
- Total addresses: 2^8 = 256
- Usable addresses: 256 - 2 = 254

8. NAT (Network Address Translation)

NAT is a method of remapping one IP address space into another by modifying network address information in IP packet headers while they pass through a routing device.

Why NAT?

Problem

- IPv4 addresses are limited (~4.3 billion)
- Every device needs a unique address on the internet
- Organizations don't have enough public IPs for all their devices

Solution: NAT

- Maps multiple private IPs to a single public IP
- Conserves scarce global IPv4 address space
- Enables devices on private networks to access the internet

How NAT Works

Process for Outgoing Traffic

1. Device on private network initiates connection
2. Device uses private IP (e.g., 192.168.1.5)
3. NAT gateway intercepts the packet
4. Gateway creates entry in translation table
5. Gateway replaces private source IP with its public IP
6. Gateway may assign a new port number
7. Modified packet is forwarded to the internet

Process for Incoming Response

1. Response arrives at NAT gateway from the internet
2. Gateway examines destination port
3. Gateway finds matching entry in translation table
4. Gateway replaces destination IP with private IP from table
5. Modified packet is forwarded to the private network device

Benefits

- Conserves IPv4 addresses
- Adds layer of security (hides internal IPs)

- Enables device mobility
- Allows network changes without affecting external addresses

Limitations

- Can complicate VPN configurations
- May cause issues with some protocols
- Adds processing overhead

9. Firewall

A **firewall** is a network security system that restricts internet traffic flowing into, out of, or within a private network based on predetermined security rules.

How Firewalls Work

Basic Process

1. Packet arrives at firewall
2. Firewall examines packet headers and contents
3. Compares against pre-set or learned rules
4. Decision: Allow, Block, or Drop
5. Forwards or rejects accordingly

Firewall Filtering Criteria

Firewalls can filter traffic based on:

Source

- Where the connection is coming from
- Source IP address

Destination

- Where the connection is intended to go
- Destination IP address

Contents

- What data is being transmitted
- Payload inspection

Packet Protocols

- Communication protocol used
- TCP/IP protocols

Application Protocols

- HTTP, FTP, DNS, Telnet, SSH
- Port-specific protocols

Firewall Rules

Example Rules

- Allow: HTTP traffic (port 80) from anywhere
- Deny: FTP traffic from external networks
- Block: All traffic from specific IP ranges
- Allow: DNS queries (port 53) to internal DNS server

What Firewalls Protect Against

Effective Protection

- **Infiltration from malicious actors:** Blocks unwanted connections and advanced persistent threats (APTs)
- **Eavesdropping:** Prevents unauthorized access to communications
- **Parental controls:** Restricts access to specific content
- **Workplace restrictions:** Controls employee internet access
- **Network segmentation:** Isolates sensitive networks

Limitations

- **Cannot identify malicious intent in legitimate traffic:** Firewalls don't understand context
- **Cannot stop internal threats:** Internal compromised devices can bypass firewalls
- **Limited against malware:** Firewalls alone cannot protect against all malware
- **Cannot stop connections bypassing the firewall:** Need internal firewalls too

Firewall Best Practices

- Regularly update firewall rules
- Implement DNSSEC for DNS security
- Use both perimeter and host-based firewalls
- Monitor and audit firewall logs
- Use multi-layer security approach

10. Port

A **network port** is a logical endpoint for network communication, identified by a number, that allows a single IP address to handle multiple simultaneous connections for different services.

Port Numbers

Range

- Total ports: 0 to 65,535
- Represented by 16-bit numbers
- Each IP address can have up to 65,535 ports

Port Identification

- Specified in URL/IP as: IP:port (e.g., 10.0.0.1:80)
- Associated with protocols: TCP, UDP, or ICMP

Port Categories

Well-Known Ports (0 - 1023)

Port	Protocol	Service
20, 21	TCP	File Transfer Protocol (FTP)
23	TCP	Telnet
25	TCP	Simple Mail Transfer Protocol (SMTP)
53	TCP/UDP	Domain Name System (DNS)
80	TCP	Hypertext Transfer Protocol (HTTP)

Registered Ports (1024 - 49,151)

- Assigned by IANA upon application
- For specific services or applications
- Can be licensed or registered for particular uses

Dynamic or Private Ports (49,152 - 65,535)

- Ephemeral or temporary ports
- Can be used by any service

- No permanent assignment
- Often used for client-side connections

Port Analogy

Think of ports like telephone extensions in a business:

- IP address = Company telephone number (gets you to the company)
- Port number = Extension (connects you to the right department/service)

Best Practices

- Use standard ports for known services
- Avoid opening unnecessary ports
- Document which ports are in use
- Monitor port activity for security threats
- Use firewalls to control port access

11. DNS (Domain Name System)

DNS is a hierarchical and distributed system that translates human-readable domain names into IP addresses (and vice versa), functioning as the internet's "phone book."

Why DNS Matters

- Users prefer remembering names (example.com) over IP addresses (93.184.216.34)
- DNS enables web browsing using domain names
- Operates transparently in the background
- Makes the internet user-friendly

DNS Record Mapping

- Domain Name → IP Address
- Example: www.google.com → 142.251.41.14

DNS Server Types

Recursive Resolver

- First DNS server consulted by users
- Usually managed by ISP or public DNS service (8.8.8.8, 1.1.1.1)
- Searches on behalf of the user
- Returns final result to user

Root Name Server

- Master index of all top-level domain servers
- Managed by ICANN
- Doesn't know the final IP address
- Directs resolver to correct TLD server

TLD (Top-Level Domain) Server

- Manages specific domain extensions (.com, .org, .edu, etc.)
- Maintains database for all domains under its extension
- Directs resolver to authoritative server

Authoritative Name Server

- Holds actual DNS records for domains
- Provides definitive answers about domain IP addresses
- Only server with true domain information
- Responds with binding between domain and IP

DNS Resolution Process (Step-by-Step)

Step 1: User Enters Domain Name

- User types "www.example.com" in browser
- Browser initiates DNS query

Step 2: Recursive Resolver Checks Cache

- Resolver checks its cache for previously resolved addresses
- If found, returns immediately
- If not found, continues to next step

Step 3: Query to Root Name Server

- If not cached, resolver contacts root name server
- Root server examines domain extension (.com)
- Directs resolver to appropriate TLD server

Step 4: Query to TLD Server

- Resolver contacts TLD server for .com domains
- TLD server provides address of authoritative server
- Narrows down the location

Step 5: Query to Authoritative Server

- Resolver contacts authoritative server for the domain
- Authoritative server holds DNS records for www.example.com
- Returns correct IP address

Step 6: IP Address Returned to Resolver

- Authoritative server sends IP address back through chain
- Resolver receives the final IP address
- Resolver caches the result for future queries

Step 7: Browser Connects to Web Server

- Browser receives IP address from resolver
- Browser connects to the website's server using IP
- Webpage loads for the user

Benefits

- Speeds up repeated domain lookups
- Reduces network traffic
- Improves browsing performance

DNS Best Practices

Security

- Implement DNSSEC (DNS Security Extensions)
- Protects against DNS spoofing attacks
- Validates DNS records authenticity

Reliability

- Use multiple DNS servers for redundancy
- Monitor DNS performance
- Configure fallback DNS servers

Performance

- Use geographically distributed DNS servers
- Implement DNS load balancing
- Cache DNS results appropriately

12. DNS Resolution Process (Detailed)

Complete DNS Query Journey

User Browser

↓

1. User types www.example.com

↓

2. Browser checks local cache

↓

3. Query sent to Recursive Resolver (usually ISP's DNS)

↓

4. Resolver checks its cache

↓

5. If not cached, query sent to Root Name Server

↓

6. Root server identifies TLD (.com) and responds with TLD server address

↓

7. Query sent to TLD Name Server for .com domains

↓

8. TLD server responds with authoritative server address

↓

9. Query sent to Authoritative Name Server (owns www.example.com records)

↓

10. Authoritative server returns IP address: 93.184.216.34

↓

11. IP address returned through resolver to browser

↓

12. Browser receives IP address and connects to web server

↓

13. Website loads for user

13. Domain Names

A **domain name** is a human-readable address used to identify a location on the internet, consisting of a name and a domain extension.

Domain Name Structure

Format: subdomain.domain.extension

Example: www.example.com

- **www** = Subdomain (optional, indicates World Wide Web)
- **example** = Domain name (main identifier)
- **.com** = Top-Level Domain extension (TLD)

Types of Domain Extensions (TLDs)

Generic Top-Level Domains (gTLDs)

- **.com** - Commercial (most common)
- **.org** - Organization (non-profit)
- **.net** - Network
- **.edu** - Educational institution
- **.gov** - Government

Country Code Top-Level Domains (ccTLDs)

- **.uk** - United Kingdom
- **.in** - India
- **.jp** - Japan
- **.de** - Germany
- **.au** - Australia

New gTLDs

- **.io** - Input/Output (tech startups)
- **.app** - Applications
- **.blog** - Blogs
- **.tech** - Technology

Domain Registration

Process

1. Choose desired domain name
2. Check availability through registrar

3. Register through domain registrar
4. Pay annual fee
5. Configure DNS records
6. Renew annually

Registrars

- Companies authorized to register domains
- Examples: GoDaddy, Namecheap, Google Domains

ICANN

- Internet Corporation for Assigned Names and Numbers
- Manages domain name system globally
- Oversees registrars

Domain vs. Subdomain

Domain

- Main website identifier (example.com)
- Can have multiple subdomains

Subdomain

- Prefix before domain name
- Examples:
    - www.example.com (www is subdomain)
    - mail.example.com (mail is subdomain)
    - blog.example.com (blog is subdomain)

Static vs. Dynamic IP with Domains

- **Static IP:** Domain always points to same IP address
- **Dynamic IP:** IP may change (use Dynamic DNS service)
- **DDNS:** Updates domain to point to dynamic IP automatically

14. Networking Commands

1. PING Command

Purpose

- Tests connectivity between computers

- Verifies if a host is reachable
- Measures response time

Syntax

text

```
ping [IP address or domain]
```

Examples

text

```
ping 8.8.8.8                # Ping IP address
ping www.google.com         # Ping domain name
ping -n 10 8.8.8.8          # Send 10 packets instead of default 4
```

Output Interpretation

- **Reply from:** Host received packet and responded
- **Bytes:** Size of data in packet
- **Time:** Round-trip time (milliseconds)
- **TTL:** Time to Live (hops remaining)
- **Request timed out:** Host unreachable or firewall blocking

Note About Firewalls

- Many firewalls block ICMP (ping requests)
- Unreachable result doesn't always mean host is down
- Could be blocked by firewall

2. IPCONFIG Command

Purpose

- Displays current IP configuration of Windows computer
- Shows network adapter settings
- Views DNS configuration

Syntax

ipconfig [options]

Common Options

text

```
ipconfig                    # Basic IP configuration
```

```
ipconfig /all              # Detailed network configuration
```

Information Displayed

- **IPv4 Address:** Current IP address
- **Subnet Mask:** Network mask
- **Default Gateway:** Router IP address
- **DHCP Server:** Server providing IP
- **DNS Servers:** DNS servers in use
- **MAC Address:** Physical address of network adapter

Examples

text

```
ipconfig                   # View basic IP info
ipconfig /all              # View comprehensive network config
```

5. NETSTAT Command

Purpose

- Displays network statistics
- Shows active connections
- Lists listening ports
- Shows network interface information

Syntax

netstat [options]

Common Options

text

```
netstat                    # Show active connections
netstat -a                 # Show all connections and listening ports
netstat -an                # Show all with numerical addresses
```

Examples

text

```
netstat -a                 # List all open ports and connections
```

netstat -an | grep LISTEN # Show listening ports (Linux)

6. ARP Command (Address Resolution Protocol)

Purpose

- Maps IP addresses to MAC addresses
- Displays and modifies ARP cache
- Troubleshoots network layer issues

Syntax

arp [options]

Common Options

text

arp -a # Display ARP cache

arp -d [IP address] # Delete ARP entry

arp -s [IP address] [MAC] # Add static ARP entry

Example Output

Interface: 192.168.1.5 --- 0x6

Internet Address Physical Address Type

192.168.1.1 00-1a-2b-3c-4d-5e dynamic

7. TELNET Command

Purpose

- Establishes remote connection to host
- Communicates with remote servers
- Accesses services on specific ports
- Troubleshoots connectivity

Syntax

telnet [IP address or hostname] [port]

Examples

text

telnet 192.168.1.1 # Connect to device

telnet www.example.com 80 # Connect to web server

```
telnet mail.example.com 25      # Test SMTP server
telnet mail.example.com 110     # Test POP3 server
```

Usage

- Useful for testing if ports are open
- Can interact with remote services
- Insecure (no encryption) - use SSH instead

Note

- TELNET sends data unencrypted
- For secure remote access, use SSH instead
- Many servers disable TELNET for security

Summary of Key Networking Concepts

Device Identification

- **IP Address:** Numerical identifier (192.168.1.1)
- **MAC Address:** Physical address (00-1a-2b-3c-4d-5e)
- **Hostname:** Computer name (mycomputer)
- **Domain:** Website address (example.com)

Network Organization

- **LAN:** Local network (office, home)
- **WAN:** Wide area network (internet)
- **Subnet:** Divided network segment
- **CIDR:** Flexible IP allocation method

Network Devices

- **Switch:** Connects devices in LAN
- **Router:** Connects different networks
- **Firewall:** Controls traffic security
- **Gateway:** Translates between networks

Services

- **DNS:** Translates domain names to IPs
- **NAT:** Translates private to public IPs

- **DHCP:** Assigns IPs automatically

Ports

- Well-known (0-1023): Standard services
- Registered (1024-49151): Application services
- Dynamic (49152-65535): Temporary use

Troubleshooting Tools

- **ping:** Test connectivity
- **ipconfig/ifconfig:** View network config
- **tracert/traceroute:** Trace packet path
- **nslookup:** Query DNS
- **netstat:** View connections
- **arp:** View IP-MAC mappings
- **telnet:** Test port connectivity